Solving System of Linear Equations

Hung-yi Lee

Reference: chapter 1, Elementary Linear Algebra - A Matrix Approach, **2nd Ed.**, by L. E. Spence, A. J. Insel and S. H. Friedberg

Outline

What is “system of linear equation”?

Solution of system of linear equation

Terminologies:

linear combination	span	dependent / independent	rank / nullity

How to find solution

“Wide matrix” must have dependent columns

System of Linear Equations

多元一次聯立方程式

coefficient

m equations

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{cases}$$

n variables

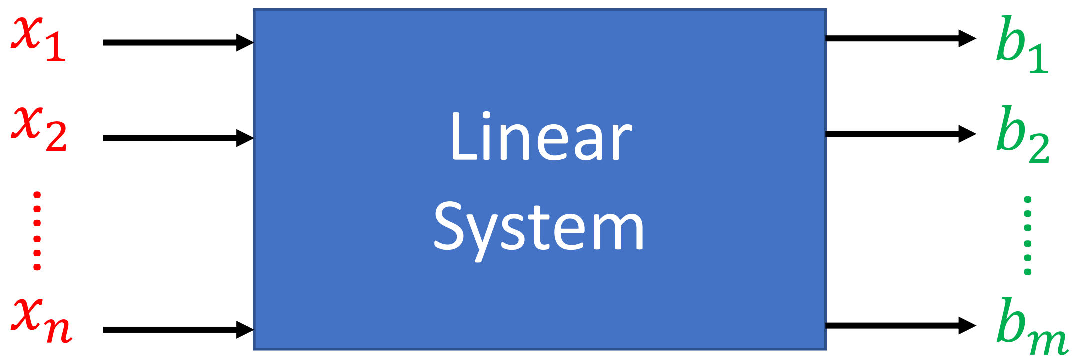
In this course, m and n can be large

System of Linear Equations

= Linear System

$$
\begin{aligned}
a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\
a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\
&\vdots \\
a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m
\end{aligned}
$$

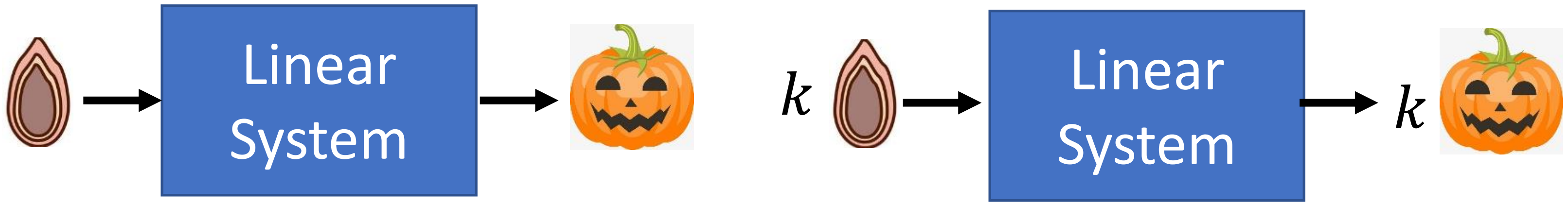
=



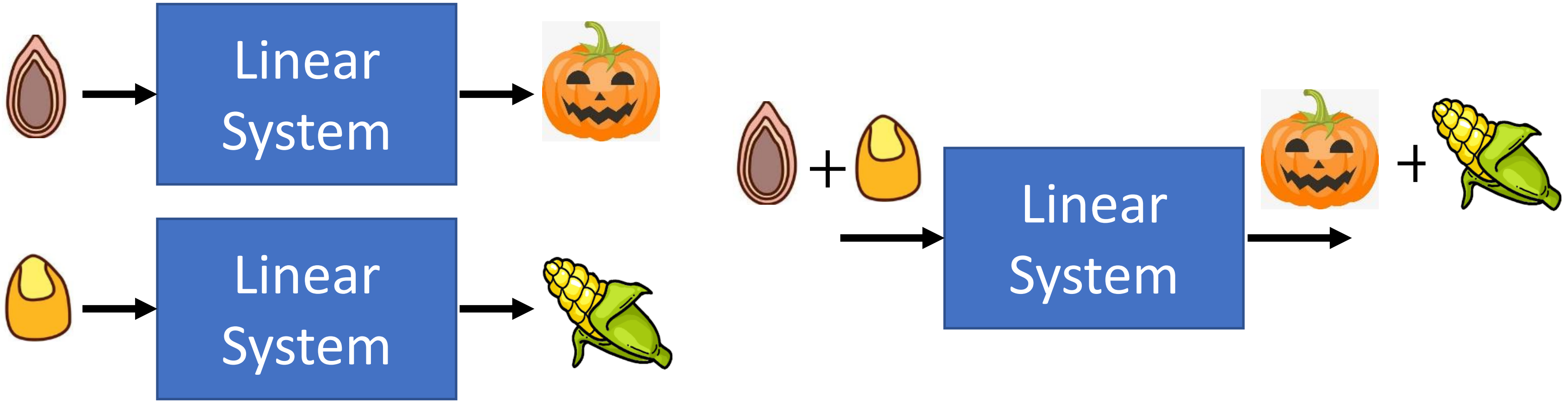
Linear System

Simple? Derivative and integral in a range are linear systems.

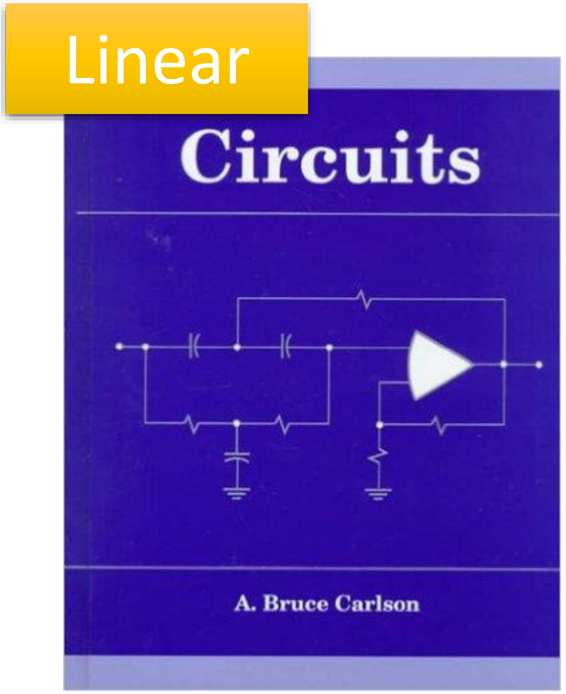
- 1. Persevering Multiplication



- 2. Persevering Addition



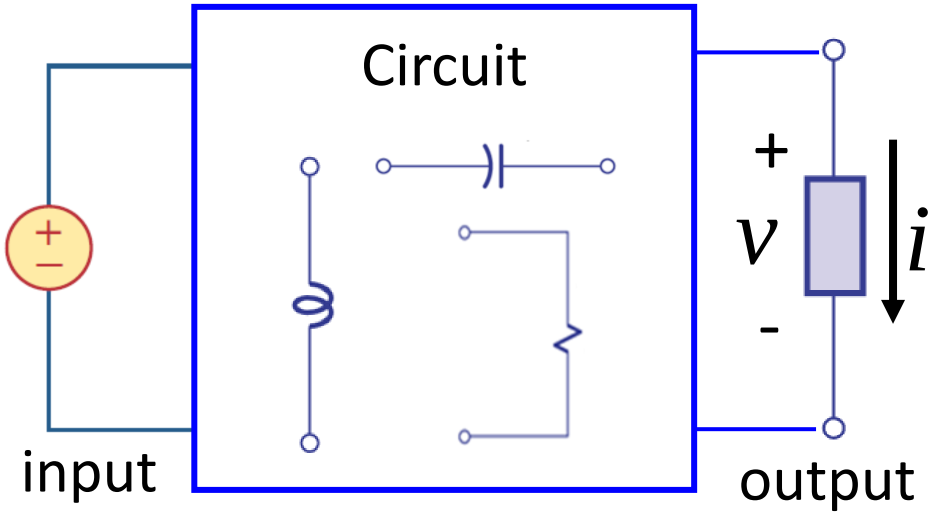
電路學



(大一必修)

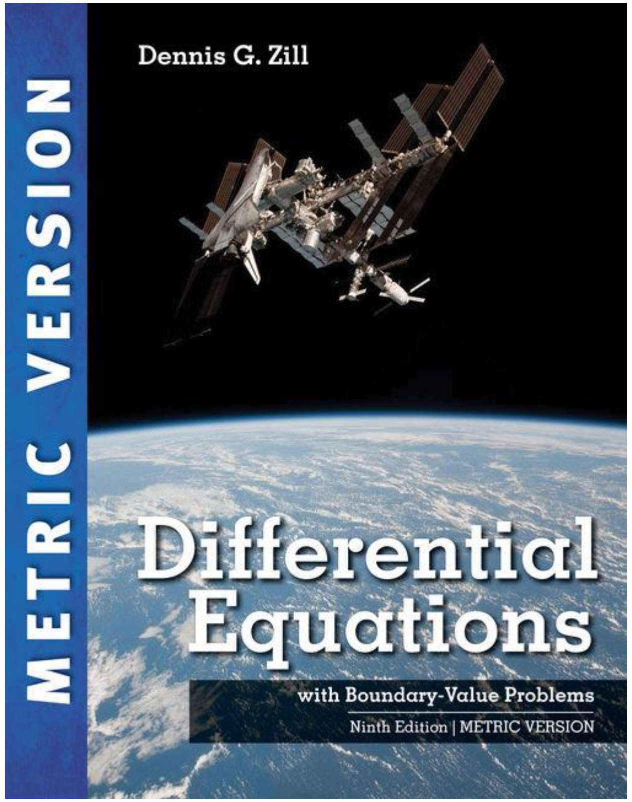
Input: voltage source, current source

output: voltage and current on the load (燈泡、引擎)

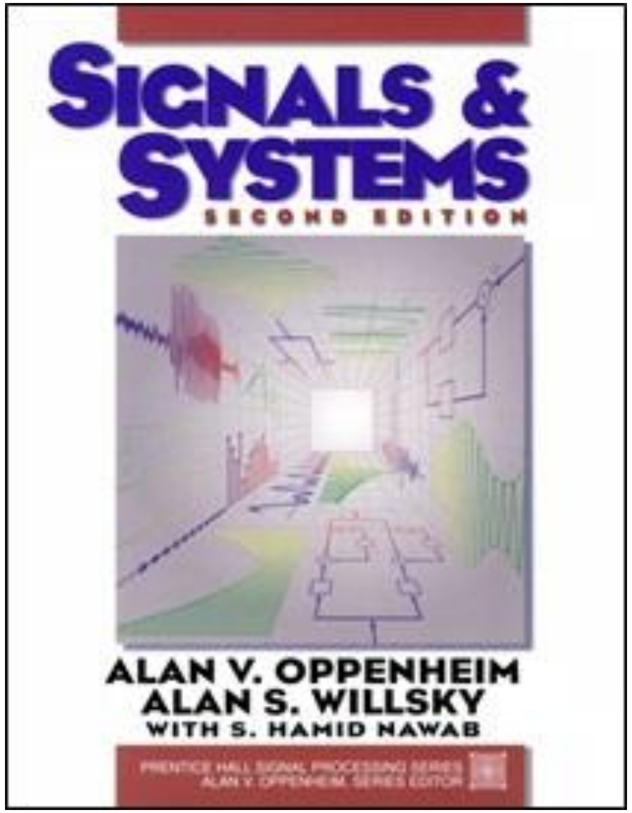


Linear System

微分方程、信號與系統



(大二上必修)



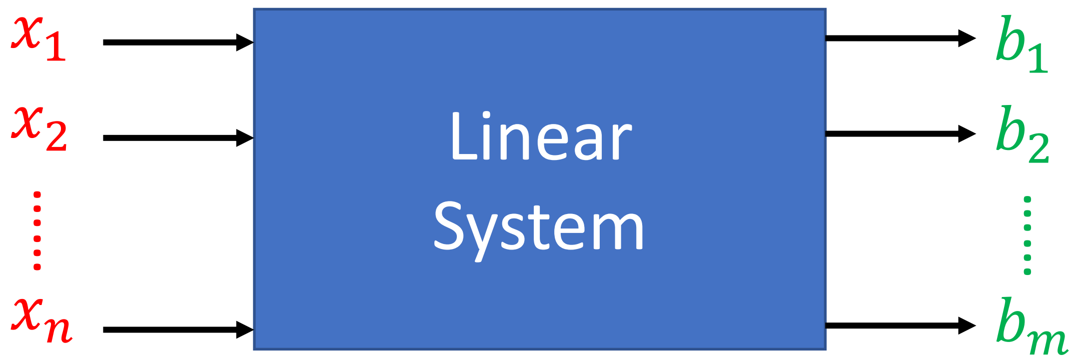
(大二下必修)

Matrix-Vector Product

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

= $A\mathbf{x} = \mathbf{b}$

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \quad \boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad \boldsymbol{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$



$\parallel$

$$
\begin{aligned}
a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\
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\end{aligned}
$$

$\parallel$

Matrix-vector product: $A\mathbf{x} = \mathbf{b}$

Outline

What is “system of linear equation”?

Solution of system of linear equation

Terminologies:

linear combination	span	dependent / independent	rank / nullity

How to find solution

“Wide matrix” must have dependent columns

Solution

$\mathbf{x} \rightarrow$ Linear System $\rightarrow \mathbf{b}$

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

System of Linear Equations

Matrix-vector product: $A\mathbf{x} = \mathbf{b}$

Given A and b, let's find x

no solution
inconsistent

unique solution
consistent

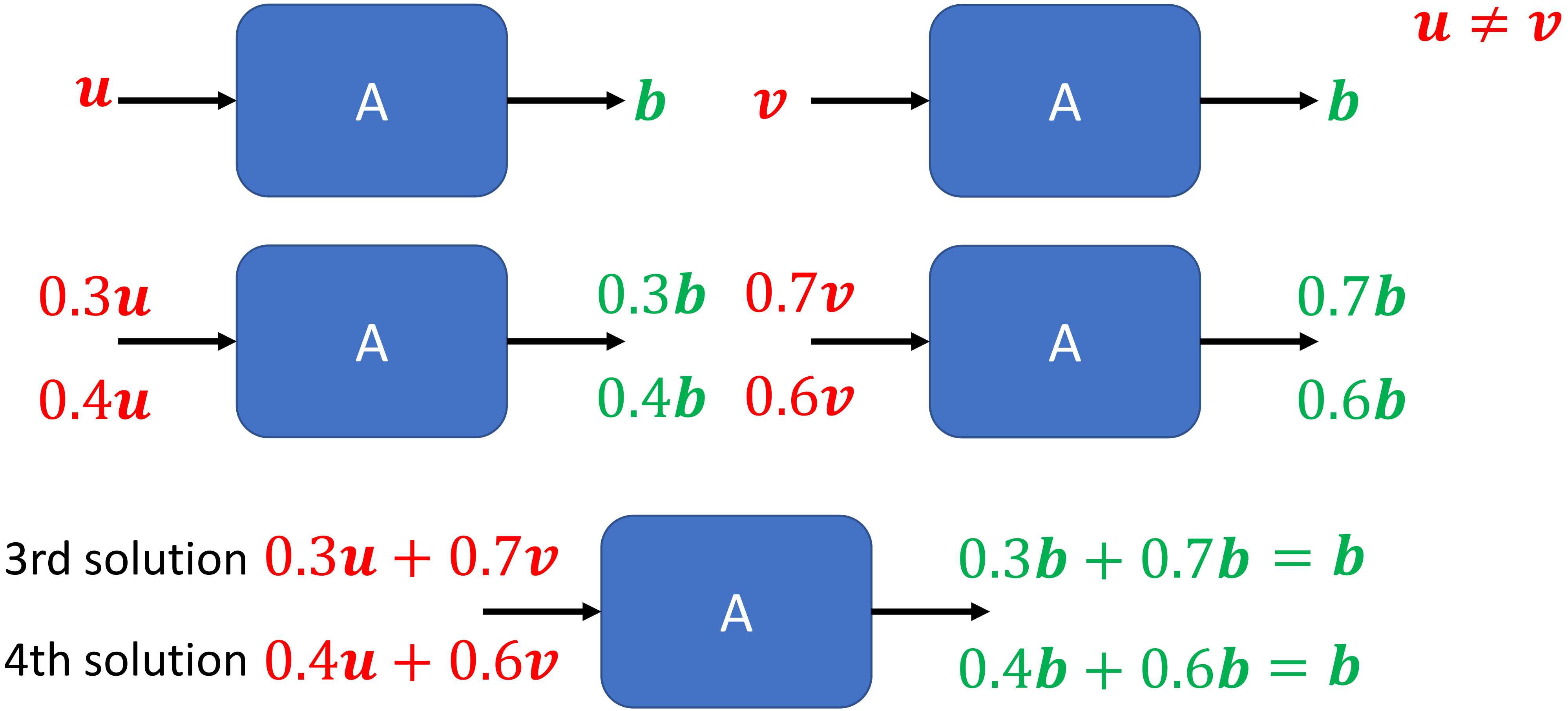
infinite solutions
consistent

為什麼不能只有兩個解？

Only Unique and Infinite?

为什麼不能只有兩個解？

一旦找到兩個解，就可以找到無窮多解



Summary

$A: m \times n \quad \boldsymbol{x} \in R^n \quad \boldsymbol{b} \in R^m$

Is $\boldsymbol{b}$ a ***linear combination*** of columns of A?

Is $\boldsymbol{b}$ in the ***span*** of the columns of A?

NO

No solution

YES

The columns of A are ***independent***.

Rank A = n

Nullity A = 0

Unique solution

The columns of A are ***dependent***.

Rank A < n

Nullity A > 0

Infinite solution

Summary

The columns of A are ***independent***.

Rank A = n

Nullity A = 0

$A: m \times n$

$\boldsymbol{x} \in R^n \quad \boldsymbol{b} \in R^m$

NO

Is $\boldsymbol{b}$ a ***linear combination*** of columns of A?

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No solution

YES

Infinite solution

YES

Is $\boldsymbol{b}$ a ***linear combination*** of columns of A?

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- span
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“Wide matrix” must have dependent columns

Linear Combination

- Given a vector set $\{\boldsymbol{u_1}, \boldsymbol{u_2}, \cdots, \boldsymbol{u_k}\}$
- Linear combination: $c_1\boldsymbol{u_1} + c_2\boldsymbol{u_2} + \cdots + c_k\boldsymbol{u_k}$
  - $c_1, c_2, \cdots, c_k$ are scalars (coefficients of linear combination)

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

$$\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

"Inner Product" with Row

$$A\boldsymbol{x} = \begin{bmatrix} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \end{bmatrix}$$

$$= x_1 \begin{bmatrix} a_{11} \\ \vdots \\ a_{m1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ \vdots \\ a_{m2} \end{bmatrix} + \cdots + x_n \begin{bmatrix} a_{1n} \\ \vdots \\ a_{mn} \end{bmatrix}$$

Linear combination of Columns

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

$$A\mathbf{x} = \mathbf{b}$$

Has solution or not?

The same question

$$x_1 \begin{bmatrix} a_{11} \\ \vdots \\ a_{m1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ \vdots \\ a_{m2} \end{bmatrix} + \cdots + x_n \begin{bmatrix} a_{1n} \\ \vdots \\ a_{mn} \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

Is $\boldsymbol{b}$ the linear combination of columns of A?

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Span

- A $\boxed{\text{vector set}}$ $S = \{\boldsymbol{u}_1, \boldsymbol{u}_2, \cdots, \boldsymbol{u}_k\}$
- $Span\ S$ is the $\boxed{\text{vector set}}$ of all linear combinations of $\boldsymbol{u}_1, \boldsymbol{u}_2, \cdots, \boldsymbol{u}_k$

$$Span\ S = \{c_1\boldsymbol{u}_1 + c_2\boldsymbol{u}_2 + \cdots + c_k\boldsymbol{u}_k \mid for\ all\ c_1, c_2, \cdots, c_k\}$$

- Vector set $V = Span\ S$
  - "S is a generating set for V" or "S generates V"
  - One way to describe a vector set with infinite elements

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

$$A\mathbf{x} = \mathbf{b}$$

Has solution or not?

The same question

$$x_1 \begin{bmatrix} a_{11} \\ \vdots \\ a_{m1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ \vdots \\ a_{m2} \end{bmatrix} + \cdots + x_n \begin{bmatrix} a_{1n} \\ \vdots \\ a_{mn} \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

Is ***b*** the linear combination of columns of A?

The same question

$$\begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix} \in Span \left\{ \begin{bmatrix} a_{11} \\ \vdots \\ a_{m1} \end{bmatrix}, \begin{bmatrix} a_{12} \\ \vdots \\ a_{m2} \end{bmatrix} \dots \begin{bmatrix} a_{1n} \\ \vdots \\ a_{mn} \end{bmatrix} \right\}$$

Is ***b*** in the span of the columns of A?

Outline

What is “system of linear equation”?

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How to find solution

“Wide matrix” must have dependent columns

依賴　　　　　　　　獨立

Dependent / Independent

- A set of n vectors $\{\boldsymbol{u}_1, \boldsymbol{u}_2, \cdots, \boldsymbol{u}_n\}$ is linear ***dependent***
  - If there exist scalars $c_1, c_2, \cdots, c_n$, **not all zero,** such that

$$c_1\boldsymbol{u}_1 + c_2\boldsymbol{u}_2 + \cdots + c_n\boldsymbol{u}_n = \mathbf{0}$$

- A set of n vectors $\{\boldsymbol{u}_1, \boldsymbol{u}_2, \cdots, \boldsymbol{u}_n\}$ is linear ***independent***

$$c_1\boldsymbol{u}_1 + c_2\boldsymbol{u}_2 + \cdots + c_n\boldsymbol{u}_n = \mathbf{0}$$

Only if $c_1 = c_2 = \cdots = c_n = 0$

Linear Dependent = Vector Set 中有成員是「依賴」的

Given a vector set, $\{\boldsymbol{u}_1, \boldsymbol{u}_2, \cdots, \boldsymbol{u}_n\}$, there exists scalars $c_1, c_2, \cdots, c_n$, that are **not all zero**, such that $c_1\boldsymbol{u}_1 + c_2\boldsymbol{u}_2 + \cdots + c_n\boldsymbol{u}_n = \boldsymbol{0}$.

(for $n \geq 2$)

=

Given a vector set, $\{\boldsymbol{u}_1, \boldsymbol{u}_2, \cdots, \boldsymbol{u}_n\}$, if there exists any $\boldsymbol{u}_i$ that is a linear combination of other vectors

$\boldsymbol{u}_i$ 是其他人的組合 ➡ $\boldsymbol{u}_i$ 是「依賴」的

Linear Independent = Vector Set 中沒有成員是「依賴」的

Dependent vs. Solution

Dependent:
Once we have solution, we have infinite.

- Intuitive link between dependence and the number of solutions

$$\begin{bmatrix} 6 & 1 & 7 \\ 3 & 8 & 11 \\ 3 & 3 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 14 \\ 22 \\ 12 \end{bmatrix}$$

dependent

$$\boxed{1 \cdot \begin{bmatrix} 6 \\ 3 \\ 3 \end{bmatrix} + 1 \cdot \begin{bmatrix} 1 \\ 8 \\ 3 \end{bmatrix} = \begin{bmatrix} 7 \\ 11 \\ 6 \end{bmatrix}}$$

$$1 \cdot \begin{bmatrix} 6 \\ 3 \\ 3 \end{bmatrix} + 1 \cdot \begin{bmatrix} 1 \\ 8 \\ 3 \end{bmatrix} + 1 \cdot \boxed{\begin{bmatrix} 7 \\ 11 \\ 6 \end{bmatrix}} = \begin{bmatrix} 14 \\ 22 \\ 12 \end{bmatrix} \qquad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$2 \cdot \begin{bmatrix} 6 \\ 3 \\ 3 \end{bmatrix} + 2 \cdot \begin{bmatrix} 1 \\ 8 \\ 3 \end{bmatrix} = \begin{bmatrix} 14 \\ 22 \\ 12 \end{bmatrix} \qquad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}$$

Infinite Solution

Outline

What is “system of linear equation”?

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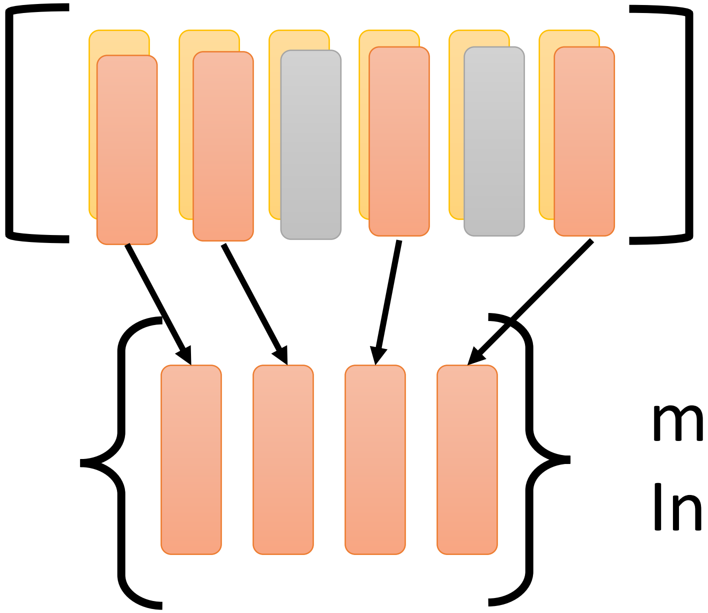
How to find solution

“Wide matrix” must have dependent columns

Rank and Nullity

- The **rank** of a matrix is defined as the maximum number of *linearly independent columns*
- **Nullity** = Number of columns - **rank**

$A =$



$Rank\ A = 4$

$Nullity\ A = 2$

maximum
Independent set

$A: m \times n$ (n columns)

The columns of A are ***independent***.

=

Rank A = n

=

Nullity A = 0

Dependent

< n

> 0

Outline

What is "system of linear equation"?

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How to find solution

"Wide matrix" must have dependent columns

$$\begin{cases} x_1 & -3x_2 & = 0 \\ 3x_1 & +x_2 & = 10 \end{cases}$$ X - 3

$$\Downarrow$$

$$\begin{cases} x_1 & -3x_2 & = 0 \\ & 10x_2 & = 10 \end{cases}$$ X 1/10

$$\Downarrow$$

$$\begin{cases} x_1 & -3x_2 & = 0 \\ & x_2 & = 1 \end{cases}$$ X 3

$$\Downarrow$$

$$\begin{cases} x_1 & & = 3 \\ & x_2 & = 1 \end{cases}$$

Equivalent

- Two systems of linear equations are **equivalent** if they have exactly **the same solution set**.

$$\begin{cases} 3x_1 & +x_2 & = 10 \\ x_1 & -3x_2 & = 0 \end{cases}$$

Solution set: $\left\{\begin{bmatrix} 3 \\ 1 \end{bmatrix}\right\}$

equivalent

$$\begin{cases} x_1 & & = 3 \\ & x_2 & = 1 \end{cases}$$

Solution set: $\left\{\begin{bmatrix} 3 \\ 1 \end{bmatrix}\right\}$

Solving system of linear equation

A ***complex*** system of linear equations

A ***simple*** system of linear equations

$$A\mathbf{x} = \mathbf{b} \;\xleftrightarrow{\text{equivalent}}\; R'\mathbf{x} = \mathbf{b}'$$

$$A' = [\, A \;\; \mathbf{b} \,] \Rightarrow A'' \Rightarrow A''' \Rightarrow \cdots\cdots \Rightarrow R = [\, R' \;\; \mathbf{b}' \,]$$

equivalent

elementary row operations

1. Interchange any two rows of the matrix
2. Multiply every entry of some row by the same nonzero scalar
3. Add a multiple of one row of the matrix to another row

$$\begin{cases} x_1 & -3x_2 & = 0 \\ 3x_1 & +x_2 & = 10 \end{cases}$$

X -3

$$\begin{bmatrix} 1 & -3 & 0 \\ 3 & 1 & 10 \end{bmatrix}$$

X -3

$$\begin{cases} x_1 & -3x_2 & = 0 \\ & 10x_2 & = 10 \end{cases}$$

X 1/10

$$\begin{bmatrix} 1 & -3 & 0 \\ 0 & 10 & 10 \end{bmatrix}$$

X 1/10

$$\begin{cases} x_1 & -3x_2 & = 0 \\ & x_2 & = 1 \end{cases}$$

X 3

$$\begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

X 3

$$\begin{cases} x_1 & & = 3 \\ & x_2 & = 1 \end{cases}$$

$$\begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 1 \end{bmatrix}$$

Solving system of linear equation

A ***complex*** system of linear equations

$A\mathbf{x} = \mathbf{b}$

?????

A ***simple*** system of linear equations

$R'\mathbf{x} = \mathbf{b}'$

equivalent

$A' = [\, A \;\; \mathbf{b} \,]$ → A'' → A''' → …… → $R = [\, R' \;\; \mathbf{b}' \,]$

Reduced Row Echelon Form (RREF)

elementary row operations:

1. Interchange any two rows of the matrix
2. Multiply every entry of some row by the same nonzero scalar
3. Add a multiple of one row of the matrix to another row

階層

Reduced Row Echelon Form

- A system of linear equations is easily solvable if its augmented matrix is in ***reduced row echelon form***
- ***Row Echelon Form (REF)***

1. Each nonzero row lies above every zero row

2. The leading entries are in echelon form

$$\begin{bmatrix} 1 & 7 & 2 & -3 & 9 & 4 \\ 0 & 0 & 1 & 4 & 6 & 8 \\ 0 & 0 & 0 & 2 & 3 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Reduced Row Echelon Form

- A system of linear equations is easily solvable if its augmented matrix is in ***reduced row echelon form***
- ***Row Echelon Form (REF)***

1. Each nonzero row lies above every zero row

2. The leading entries are in echelon form

NO

$$\begin{bmatrix} 1 & 0 & 0 & 6 & 3 & 0 \\ 0 & 0 & 1 & 5 & 7 & 0 \\ 0 & 1 & 0 & 2 & 4 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

No zero rows

Reduced Row Echelon Form

- A system of linear equations is easily solvable if its augmented matrix is in ***reduced row echelon form***
- ***Reduced Row Echelon Form (RREF)***

1-2 The matrix is in row echelon form

3. The columns containing the leading entries are standard vectors.

$$\begin{bmatrix} 1 & 7 & 2 & -3 & 9 & 4 \\ 0 & 0 & 1 & 4 & 6 & 8 \\ 0 & 0 & 0 & 2 & 3 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

只有一個 1 ，其餘都為 0

Reduced Row Echelon Form

- A system of linear equations is easily solvable if its augmented matrix is in ***reduced row echelon form***
- ***Reduced Row Echelon Form (RREF)***

1-2 The matrix is in row echelon form

3. The columns containing the leading entries are standard vectors.

$$\begin{bmatrix} 1 & -3 & 0 & 2 & 0 & 7 \\ 0 & 0 & 1 & 6 & 0 & 9 \\ 0 & 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Reduced Row Echelon Form

- A system of linear equations is easily solvable if its augmented matrix is in ***reduced row echelon form***

Example 1. Unique Solution

$$\begin{array}{cccc} x_1 & x_2 & x_3 & b \\ \end{array}$$

$$\left[\begin{array}{cccc} 1 & 0 & 0 & -4 \\ 0 & 1 & 0 & -5 \\ 0 & 0 & 1 & 3 \end{array}\right] \Rightarrow \begin{aligned} x_1 \quad\quad &= -4 \\ x_2 \quad &= -5 \\ x_3 &= \ \ 3 \end{aligned}$$

If RREF looks like [I $\mathbf{b}'$]

unique solution

Example 2. Infinite Solution

$$\begin{array}{cccccc} x_1 & x_2 & x_3 & x_4 & x_5 & b \\ \end{array}$$

$$\left[\begin{array}{cccccc} 1 & -3 & 0 & 2 & 0 & 7 \\ 0 & 0 & 1 & 6 & 0 & 9 \\ 0 & 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

$$\begin{aligned} \boxed{x_1} - 3x_2 + 2x_4 &= 7 \\ \boxed{x_3} + 6x_4 &= 9 \\ x_5 &= 2 \\ \cancel{0 = 0} \end{aligned}$$

Free variables

Basic variables

With free variables, there are infinitely many solutions.

Parametric Representation:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 7 + 3x_2 - 2x_4 \\ \\ 9 - 6x_4 \\ \\ 2 \end{bmatrix}$$

Reduced Row Echelon Form

- ***<u>Example 3. No Solution</u>***

$$\begin{array}{cccc} x_1 & x_2 & x_3 & b \\ \end{array}$$

$$\left[\begin{array}{cccc} 1 & 0 & -3 & 0 \\ 0 & 1 & 2 & 0 \\ \hline 0 & 0 & 0 & 1 \\ \hline 0 & 0 & 0 & 0 \end{array}\right] \Leftrightarrow \begin{aligned} x_1 \qquad\quad -3x_3 &= 0 \\ x_2 + 2x_3 &= 0 \\ 0x_1 + 0x_2 + 0x_3 &= 1 \\ 0x_1 + 0x_2 + 0x_3 &= 0 \end{aligned}$$

inconsistent

When an augmented matrix contains a row in which **the only nonzero entry lies in the last column**

$\Rightarrow$ The corresponding system of linear equations has **no solution** (**inconsistent**).

Outline

What is “system of linear equation”?

Solution of system of linear equation

Terminologies:

linear combination	span	dependent / independent	rank / nullity

How to find solution

“Wide matrix” must have dependent columns

"Wide matrix" must be dependent

$$\begin{bmatrix} 1 & 1 \\ 2 & 2 \\ 3 & 3 \end{bmatrix}$$

dependent

$$\begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix} \quad \begin{matrix} \Longrightarrow \\ \not\Longleftarrow \end{matrix} \quad \text{The columns are dependent}$$

(wide matrix)

- **More than 3 vectors in R^3 must be dependent.**

$$\begin{bmatrix} * \\ * \\ * \end{bmatrix} \begin{bmatrix} * \\ * \\ * \end{bmatrix} \begin{bmatrix} * \\ * \\ * \end{bmatrix} \begin{bmatrix} * \\ * \\ * \end{bmatrix} \Rightarrow \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix}$$

- **More than m vectors in R^m must be dependent.**

“Wide matrix” must be dependent

Linear combination of the other three

$$\underbrace{\begin{bmatrix}*\\*\\*\end{bmatrix}\begin{bmatrix}*\\*\\*\end{bmatrix}\begin{bmatrix}*\\*\\*\end{bmatrix}}_{\text{Independent}}\begin{bmatrix}*\\*\\*\end{bmatrix}$$

?

$$\underbrace{\begin{bmatrix}*\\*\\*\end{bmatrix}\begin{bmatrix}*\\*\\*\end{bmatrix}\begin{bmatrix}*\\*\\*\end{bmatrix}}_{\text{Already dependent}}\begin{bmatrix}*\\*\\*\end{bmatrix}$$

$A: 3 \times 3$ $\quad$ $\boldsymbol{x} \in R^3$ $\quad$ $\boldsymbol{b} \in R^3$

$A\mathbf{x} = \mathbf{b}$ is consistent for ***every*** $\boldsymbol{b}$

Every $\boldsymbol{b}$ is in the span of the columns of

$A = [\boldsymbol{a}_1 \quad \boldsymbol{a}_2 \quad \boldsymbol{a}_3]$

$Span\underbrace{\{\boldsymbol{a}_1, \boldsymbol{a}_2, \boldsymbol{a}_3\}}_{\text{Independent}} = R^3$ $\quad$ 這就是為什麼 “三人行必有我師”

m independent vectors can span R^m

Pivot 中樞

Pivot Positions and Pivot Columns

$$A = \begin{bmatrix} 1 & 2 & -1 & 2 & 1 & 2 \\ -1 & -2 & 1 & 2 & 3 & 6 \\ 2 & 4 & -3 & 2 & 0 & 3 \\ -3 & -6 & 2 & 0 & 3 & 9 \end{bmatrix} \Rightarrow R = \begin{bmatrix} 1 & 2 & 0 & 0 & -1 & -5 \\ 0 & 0 & 1 & 0 & 0 & -3 \\ 0 & 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Leading Entry

The pivot positions of A are (1,1), (2,3) and (3,4).

The pivot columns of A are 1st, 3rd and 4th columns.

Column Correspondence Theorem

$$A = [\boldsymbol{a}_1 \quad \cdots \quad \boldsymbol{a}_n] \xrightarrow{\text{RREF}} R = [\boldsymbol{r}_1 \quad \cdots \quad \boldsymbol{r}_n]$$

If $\boldsymbol{a}_j$ is a linear combination of other columns of A → $\boldsymbol{r}_j$ is a linear combination of the ***corresponding*** columns of R with ***the same coefficients***

$\mathbf{a}_5 = -\mathbf{a}_1 + \mathbf{a}_4$ $\qquad$ $\mathbf{r}_5 = -\mathbf{r}_1 + \mathbf{r}_4$

$\boldsymbol{a}_j$ is a linear combination of the ***corresponding*** columns of A with ***the same coefficients*** ← If $\boldsymbol{r}_j$ is a linear combination of other columns of R

$\mathbf{a}_3 = 3\mathbf{a}_1 - 2\mathbf{a}_2$ $\qquad$ $\mathbf{r}_3 = 3\mathbf{r}_1 - 2\mathbf{r}_2$

Column Correspondence Theorem - Example

$$A = \begin{array}{c} \begin{array}{cccccc} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 & \mathbf{a}_4 & \mathbf{a}_5 & \mathbf{a}_6 \end{array} \\ \begin{bmatrix} 1 & 2 & -1 & 2 & 1 & 2 \\ -1 & -2 & 1 & 2 & 3 & 6 \\ 2 & 4 & -3 & 2 & 0 & 3 \\ -3 & -6 & 2 & 0 & 3 & 9 \end{bmatrix} \end{array} \qquad R = \begin{array}{c} \begin{array}{cccccc} \mathbf{r}_1 & \mathbf{r}_2 & \mathbf{r}_3 & \mathbf{r}_4 & \mathbf{r}_5 & \mathbf{r}_6 \end{array} \\ \begin{bmatrix} 1 & 2 & 0 & 0 & -1 & -5 \\ 0 & 0 & 1 & 0 & 0 & -3 \\ 0 & 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

$$\mathbf{a}_2 = 2\mathbf{a}_1 \qquad \Longleftrightarrow \qquad \mathbf{r}_2 = 2\mathbf{r}_1$$

Column Correspondence Theorem - Example

$$A = \begin{array}{c} \begin{matrix} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 & \mathbf{a}_4 & \mathbf{a}_5 & \mathbf{a}_6 \end{matrix} \\ \begin{bmatrix} 1 & 2 & -1 & 2 & 1 & 2 \\ -1 & -2 & 1 & 2 & 3 & 6 \\ 2 & 4 & -3 & 2 & 0 & 3 \\ -3 & -6 & 2 & 0 & 3 & 9 \end{bmatrix} \end{array} \qquad R = \begin{array}{c} \begin{matrix} \mathbf{r}_1 & \mathbf{r}_2 & \mathbf{r}_3 & \mathbf{r}_4 & \mathbf{r}_5 & \mathbf{r}_6 \end{matrix} \\ \begin{bmatrix} 1 & 2 & 0 & 0 & -1 & -5 \\ 0 & 0 & 1 & 0 & 0 & -3 \\ 0 & 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

$$\mathbf{a}_2 = 2\mathbf{a}_1 \quad \Longleftrightarrow \quad \mathbf{r}_2 = 2\mathbf{r}_1$$

$$\mathbf{a}_5 = -\mathbf{a}_1 + \mathbf{a}_4 \quad \Longleftrightarrow \quad \mathbf{r}_5 = -\mathbf{r}_1 + \mathbf{r}_4$$

Column Correspondence Theorem (Column 間的承諾)

$R = [\boldsymbol{r}_1 \quad \cdots \quad \boldsymbol{r}_n]$

$A = [\boldsymbol{a}_1 \quad \cdots \quad \boldsymbol{a}_n]$

$\boldsymbol{r}_j$ is a linear combination of the ***corresponding*** columns

If $\boldsymbol{a}_j$ is a linear combination of other columns of A

Source of image: http://feetsprint.blogspot.com/2016/02/blog-post_29.html

Column Correspondence Theorem

pivot columns

$$A = \begin{bmatrix} \boxed{1} & 2 & \boxed{-1} & \boxed{2} & 1 & 2 \\ -1 & -2 & 1 & 2 & 3 & 6 \\ 2 & 4 & -3 & 2 & 0 & 3 \\ -3 & -6 & 2 & 0 & 3 & 9 \end{bmatrix}$$

linear independent

Leading entries

$$R = \begin{bmatrix} 1 & 2 & 0 & 0 & -1 & -5 \\ 0 & 0 & 1 & 0 & 0 & -3 \\ 0 & 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

linear independent

The pivot columns are linear independent.

Column Correspondence Theorem

pivot columns

Leading entries

$$A = \begin{bmatrix} 1 & 2 & -1 & 2 & 1 & 2 \\ -1 & -2 & 1 & 2 & 3 & 6 \\ 2 & 4 & -3 & 2 & 0 & 3 \\ -3 & -6 & 2 & 0 & 3 & 9 \end{bmatrix} \qquad R = \begin{bmatrix} 1 & 2 & 0 & 0 & -1 & -5 \\ 0 & 0 & 1 & 0 & 0 & -3 \\ 0 & 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$\mathbf{a}_2 = 2\mathbf{a}_1$

$\mathbf{a}_5 = -\mathbf{a}_1 + \mathbf{a}_4$

$\mathbf{a}_6 = -5\mathbf{a}_1 - 3\mathbf{a}_3 + 2\mathbf{a}_4$

$\mathbf{r}_2 = 2\mathbf{r}_1$

$\mathbf{r}_5 = -\mathbf{r}_1 + \mathbf{r}_4$

$\mathbf{r}_6 = -5\mathbf{r}_1 - 3\mathbf{r}_3 + 2\mathbf{r}_4$

The non-pivot columns are the linear combination of the previous pivot columns.

“Wide matrix” must be dependent

$$\begin{array}{cc} & n \\ m & \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix} \\ & A \end{array}$$

$$\underbrace{\hphantom{* \quad * \quad * \quad *}}_{\text{Dependent}}$$

There are at most m pivot columns.

$$m = 3 \quad \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

If $m < n$

There are non-pivot columns.

Linear combination of pivot columns

Outline

What is "system of linear equation"?

Solution of system of linear equation

Terminologies:

linear combination	span	dependent / independent	rank / nullity

How to find solution

"Wide matrix" must have dependent columns